

Your grade: 92%

Your latest: 92% • Your highest: 92% • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed skill in understanding Git commands, recommender systems, and GANs, but there is room for growth in implementing autoencoders. Before retrying this assignment, we recommend focusing on this key area:

- **Autoencoders:** In Question 3, your understanding of the steps involved in implementing an autoencoder using PyTorch was incomplete. Try reviewing [Autoencoders / Coding](#) to refresh your memory on defining encoder and decoder classes and other necessary steps.

Good job.

Retry assignment



1. Which command would you use to clone the GitHub repository containing the supporting files for the course?

1 / 1 point

- ☒ git clone https://github.com/example/repo.git
- ☐ git fetch https://github.com/example/repo.git
- ☐ git copy https://github.com/example/repo.git
- ☐ git pull https://github.com/example/repo.git

✔ Correct
Correct! This is the standard command to clone a GitHub repository.

2. In the context of recommender systems, what is the primary purpose of using Matrix Factorization?

1 / 1 point

- ☐ To encode categorical variables into numerical values
- ☐ To preprocess movie ratings before training a recommender system
- ☐ To compute the precision at K and recall at K metrics
- ☒ To decompose the user-item interaction matrix into lower-dimensional matrices

✔ Correct
Correct! Matrix Factorization decomposes the user-item interaction matrix to discover latent features.

3. Which of the following steps are involved in implementing an autoencoder using PyTorch?

0.6 / 1 point

- ☐ Initialize the autoencoder model
- ☒ Define encoder and decoder classes

✔ Correct
Correct! You need to define the encoder and decoder classes as part of the autoencoder implementation.

- ☐ Select an appropriate optimization algorithm
- ☐ Generate random noise
- ☐ Perform data encryption

You didn't select all the correct answers

4. What steps are involved in setting up classes for the discriminator and generator in a GAN?

1 / 1 point

- ☐ Data augmentation
- ☒ Creating the loss functions

✔ Correct
Correct. Loss functions are crucial for training both the generator and the discriminator.

- ☐ Preprocessing the input data
- ☒ Defining the network architecture

✔ Correct
Correct. Defining the architecture is essential for both the generator and discriminator.

- ☒ Implementing forward propagation

✔ Correct
Correct. Forward propagation must be implemented to pass data through the network.

5. Which of the following best describes the process of implementing a GAN for training a generator to create a specific shape?

1 / 1 point

- ☐ Using reinforcement learning to train an agent to create shapes based on rewards
- ☒ Setting up the generator and discriminator networks, defining loss functions, and iteratively training both networks
- ☐ Applying data augmentation techniques to expand the dataset and training a convolutional neural network
- ☐ Collecting a large labeled dataset, preprocessing the data, and training a supervised learning model

✔ Correct
Correct! Implementing a GAN involves setting up both networks, defining loss functions, and iteratively training them.